

Triple Products of Eigenfunctions and Spectral Geometry

Joe Schaefer
President, SunStar Systems
joe@sunstarsys.com

2026
July

Abstract

Using elementary techniques from Geometric Analysis, Partial Differential Equations, and Abelian C^* Algebras, we uncover a novel, yet familiar, global geometric invariant – namely the indexed set of integrals of triple products of eigenfunctions of the Laplace-Beltrami operator, to precisely characterize which isospectral closed Riemannian manifolds are isometric.

Keywords: spectrum, Fourier, harmonic, triple products, Laplacian, eigenfunctions, functional, inverse problem, representation theory

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1 Introduction

For a closed Riemannian manifold (M, g) , characterizing its class of non-isometric, isospectral manifolds is a type of Inverse Problem [DH11] in Spectral Geometry. Naïvely one might speculate that this class would always be empty. However, the academic literature is rich with decades-old constructions of specific pairings of counterexamples: beginning in 1964 with John Milnor’s 16-dimensional pair of non-isometric, isospectral flat tori [Mil64], and continuing [CS92] towards the generic dimensional characterization of flat tori in Alexander Schieman’s 1993 doctoral thesis [Sch93] – replete with a computer aided search for the critical $\dim = 3$ case. A modern survey of the full flat tori history appears in [NRR23].

Along the way were insightful offshoots into more sophisticated, non-Euclidean symmetric covering spaces; constructing such isospectral, non-isometric “duets” involving nontrivial curvature tensors (and their spectrum-determined Euler characteristics in dimension 2 [MS67].) A prime example of this effort was Toshikazu Sunada’s 1985 [Sun85] invention of a general-purpose covering space framework, which he then deployed in the same work to construct hyperbolic duets in dimensions 2 and 3.

For inhomogeneous Riemannian metrics, Carolyn Gordon discovered duets that are not even locally isometric [Gor93].

Work continues in many related areas [DH11], such as determining topological characteristics of the class of isospectral, non-isometric manifolds in general (empty [Tan80], finite [Sch93], rigid [GK80], and compact [Zho97]) as a subset of different moduli spaces of Riemannian metrics.

What we offer in this article is a new perspective on a familiar tool: indexed Fourier coefficients of pairwise products of eigenfunctions as a discrete "algebraic/topological invariant" to complement the existing, discrete "analytic invariant" – the non-negative spectrum of the Laplace-Beltrami operator (herein referred to as the (non-negative) *Laplacian*) on $\mathcal{H} = L^2(M, g)$. Combined, we observe the pair provides a "discrete global geometric representation" of the isometry classes of isospectral, closed Riemannian manifolds.

1.1 Results

Theorem 1. *Given a (non-decreasing on the eigenvalues) orthonormal basis of eigenfunctions $\{e^i\}_{i=0}^\infty$ for the (non-negative) Laplacian Δ_M on $L^2(M, g)$ associated with a closed Riemannian manifold (M, g) , define*

$$M^{i,j,k} := \int_M e^i e^j \bar{e}^k \sqrt{g} dx = \langle e^i e^j | e^k \rangle \quad (1)$$

To be isometric to (M, g) , it is a necessary and sufficient condition for another isospectral closed Riemannian manifold to have an orthonormal basis of eigenfunctions (for its Laplacian) that both preserves the associated eigenvalues and possesses an invariant $\{M^{i,j,k}\}$ under each basis.

Symmetry plays an important role in computationally tractable cases [Fra23] [LS18] [Sar94], which is aptly illustrated in our flat tori Example 5 below. However, the strength of our approach is perhaps best made apparent in the case of manifolds with the fewest number of Riemannian symmetries, which is the generic case. In this instance, we offer the following

Corollary 2. *(Diagonal Litmus Test) Given a pair of eigenvalue preserving orthonormal bases as described in the hypothesis of the Theorem, the manifolds are isometric if for every $k > 0$, there is at least one value of $i > 0$ for which $M^{i,i,k} \neq 0$ in both bases; if for every choice of i, j, k , the product $M^{i,i,k} \bar{M}^{j,j,k}$ agrees in both bases; and if the vector space spanned by $\{|e^i|^2\}$ separates points in either basis. Here $\bar{j}$ represents the eigenfunction $\bar{e}^j$ in the triple-product integral computations.*

Corollary 3. *Generically, isospectral manifolds are isometric if and only if the products as defined in Corollary 2 agree.*

The motivation for the study of $\{M^{i,j,k}\}$ is loosely derived from the study of the role of the bilinear multiplication operator $Y : V \otimes V \rightarrow V((z))$ in the definition of a Vertex Operator Algebra [FB04] associated with a Chiral Conformal Field Theory. Here V is the Vector Space of States and $V((z))$ is the space of formal Laurent series in z with coefficients in V . Since V often comes equipped as a Hilbert Space with a traditional Fourier series orthonormal basis, indexing Y using the Fourier basis elements of V is only slightly more involved than the $M^{i,j,k}$ case studied here, but quite similar in spirit. However a detailed comparison is out of scope for this article.

If we consider the map

$$(M, g, \{e^i\}) \mapsto \{\lambda_i, M^{i,j,k}\} \quad , \quad (2)$$

this paper establishes the injectivity of this map for closed Riemannian manifolds (up to Riemannian isometry in its domain). Further results which apply these techniques to describe its image (and inverse), within select moduli spaces of metrics, are just getting started [Adv25]. There, Anshul Advé rigorously tackles unit tangent spaces of compact, hyperbolic 2-orbifolds, using these same *structure constants* from Conformal Field Theory.

The generic Riemannian metric case is completely characterized by the study of $\{\lambda_i, M^{i,i,k}\}$ in the Corollaries.

These results were first demonstrated during a similarly titled talk by the author at MSRI in 1997, but they appear here in published form for the first time.

2 Preliminaries

Now with $M, g, e^i, M^{i,j,k}$ as in the hypotheses of Theorem 1, for $f \in C^\infty(M)$ and $i \geq 0$ note that the *Fourier coefficients*

$$\begin{aligned} \hat{f}(i) &:= \int_M f(x) \bar{e}^i(x) \sqrt{g(x)} dx \\ \implies \\ f(x) &= \sum_{i=0}^{\infty} \hat{f}(i) e^i(x), \end{aligned} \tag{3}$$

since f is uniquely representable as its rapidly converging *Fourier Series* (Δ_M -specific Sobolev Embeddings [Tay13] [RS75], together with Weyl's Asymptotic Law [Wey11], imply the terms in the sum are $o(i^{-n})$ uniformly in x [Hör68], $\forall n \in \mathbb{N}$.) Then we see that for $f_1, f_2 \in C^\infty(M)$, the Fourier coefficients of the pointwise product $f_1 f_2 \in C^\infty(M)$ are

$$\begin{aligned} \widehat{f_1 f_2}(k) &= \sum_{i,j}^{\infty} \hat{f}_1(i) \hat{f}_2(j) M^{i,j,k} \\ \implies \\ f_1 f_2(x) &= \sum_{i,j,k} \hat{f}_1(i) \hat{f}_2(j) M^{i,j,k} e^k(x) \\ f_1 &= f_2^p, p > 2 \implies \\ \sum_k \hat{f}_1(k) e^k(x) &= \sum_{i_1, i_2, \dots, i_{2p-1}} \hat{f}_2(i_1) \hat{f}_2(i_2) \hat{f}_2(i_4) \hat{f}_2(i_6) \dots \hat{f}_2(i_{2p-2}) M^{i_1, i_2, i_3} M^{i_3, i_4, i_5} \dots M^{i_{2p-3}, i_{2p-2}, i_{2p-1}} e^{i_{2p-1}}(x) \end{aligned} \tag{4}$$

and so, critically, any multivariate polynomial $\wp \in \mathbb{C}[z_1, \dots, z_l]$ (on smooth functions) commutes with any spectrum-preserving Δ -eigenfunction orthonormal basis map $\vec{F}$ that preserves $\{M^{i,j,k}\}$:

$$\begin{array}{ccc} C^\infty(M, \mathbb{C}^l) & \xrightarrow{\wp} & C^\infty(M) \\ \downarrow \begin{smallmatrix} \vec{F} \oplus \dots \oplus \vec{F} \\ l \text{ times} \end{smallmatrix} & & \downarrow \vec{F} \\ C^\infty(N, \mathbb{C}^l) & \xrightarrow{\wp} & C^\infty(N) \end{array} \tag{5}$$

Moreover if $A \subset M$ is Borel-measurable, then the results above hold pointwise for the characteristic function of A everywhere except along the boundary of A : if $f = f^2$ and $A := \{x \in M \mid f(x) = 1\}$,

$$\sum_i \hat{f}(i) e^i(x) = \sum_{i,j,k} \hat{f}(i) \hat{f}(j) M^{i,j,k} e^k(x) = \begin{cases} 1 & x \in \mathring{A} \\ 0 & x \in \mathring{A}^c \end{cases}, \tag{6}$$

and by uniqueness, we have the following identity

$$\begin{aligned} \hat{f}(k) &= \sum_{i,j} \hat{f}(i) \hat{f}(j) M^{i,j,k}, \forall k \geq 0 \\ \iff f &= f^2 \text{ a.e.} \end{aligned} \tag{7}$$

This implies any such basis map as above carries characteristic functions (as members of $L^2(M, g) \subset L^1(M, g)$) to characteristic functions in a measure-preserving fashion.

The point of these computations is to emphasize the fact that $\{M^{i,j,k}\}$ characterizes the Harmonic Analysis of the pointwise multiplication operator on $C^\infty(M)$, which is a dense subalgebra of the Abelian C^* algebra $C(M)$, by the Stone-Weierstrass theorem.

For the rapid convergence of these above sums involving $M^{i,j,k}$, note that products of eigenfunctions are smooth, so these Fourier coefficients decay as above (in each index). For more details, see Emmett Wyman's work in 2022 with these coefficients as it relates to the triangle inequality on the eigenvalues [Wym22].

Note: we may always assume

$$\begin{aligned}
e^0 &= M^{0,0,0} = 1/\sqrt{\text{vol}(M)} \\
&\implies \\
M^{0,j,k} &= M^{j,0,k} = \delta_{j-k}/\sqrt{\text{vol}(M)},
\end{aligned} \tag{8}$$

where δ_i is the Kronecker delta. Since $\text{vol}(M)$ is a spectral invariant [Wey11], this information is already available from isospectrality considerations.

3 Proof of Theorem 1

Proof. For necessity, let $F : (N, h) \rightarrow (M, g)$ be an isometry between closed Riemannian manifolds, and let the target orthonormal basis of eigenfunctions on $L^2(N, h)$ be the pull-back via F of the orthonormal basis $\{e^i\}$ on (M, g) above. Since

$$\begin{aligned}
M^{i,j,k} &= \int_M e^i e^j \bar{e}^k \sqrt{g} dy \\
&= \int_N e^i(F(x)) e^j(F(x)) \bar{e}^k(F(x)) \sqrt{h} dx,
\end{aligned} \tag{9}$$

we are done with the necessity argument because $\Delta_N(f \circ F) = (\Delta_M f) \circ F$, $\forall f \in C^\infty(M)$.

For sufficiency, we now consider the linear, bijective orthonormal eigenfunction basis map $\vec{F}$ from $C^\infty(M)$ to $C^\infty(N)$ and note that from the calculations in Section 2 above, $\vec{F}$ preserves pointwise products for smooth functions (and preserves characteristic functions when extended to $L^2(M, g)$) by the premise that $\{M^{i,j,k}\}$ is invariant under this map.

Lemma 4. $\vec{F} : C^\infty(M) \rightarrow C^\infty(N)$ preserves the uniform norm.

Proof of Lemma 4. Let $\{a_i\}$ be a smooth partition of unity on M .

$$\begin{aligned}
1 &= \sum_i a_i(x) \\
&= \sum_{i,j} \hat{a}_i(j) e^j(x) \\
&= \sum_j e^j(x) \sum_i \hat{a}_i(j)
\end{aligned} \tag{10}$$

Thus $\sum_i \hat{a}_i(j) = \delta_j \sqrt{\text{vol}(M)}$ (Kronecker delta).

By the dominated convergence theorem,

$$\lim_{p \rightarrow \infty} \sum_j \hat{a}_j^p(k) = \int_{\bigcup_j \{a_j=1\}} \bar{e}^k(x) \sqrt{g} dx, \tag{11}$$

which is a characteristic function of positive measure on each disjoint subset $\{x \in M \mid a_j(x) = 1\}$. This means the Lemma is proven for each a_j , since the limiting characteristic function of a set with positive measure is preserved, and hence has uniform norm 1, as do all a_j^p , $\vec{F}(a_j^p) = \vec{F}(a_j)^p$, $p \in \mathbb{N}$, by Diagram 5.

Without loss of generality, we may apply the special case result shown for the smooth partition of unity $\{|f|/\|f\|_\infty, 1 - |f|/\|f\|_\infty\}$, where $\{x \in M \mid |f(x)| = \|f\|_\infty\}$ has positive measure, and the Lemma is proven in full. ■

Since $\{\bar{e}^i\}$ is also a Fourier basis for $L^2(M, g)$, it is clear from Equation 3 that $\vec{F}(\bar{f}) = \bar{\vec{F}}(f)$. This means that on a dense set of $C(M)$ (and $C(N)$), we have established $\vec{F}$ as an isomorphism of Abelian C^* algebras, and thus can be extended to an isomorphism of $C(M)$ and $C(N)$ in the same category.

Now we apply the Gelfand-Naimark Representation Theorem (in contravariant functor form) for unital Abelian C^* algebras [Con19] to represent this isomorphism $\vec{F}$ by a homeomorphism F between N and M . Since $\vec{F}$ is bijective on smooth functions, F too must be smooth.

As this now diffeomorphism preserves eigenvalues and eigenfunctions (by hypothesis on $\vec{F}(f) = f \circ F$), it must preserve the Laplacian on smooth functions. Hence F also must preserve the principal symbols of these same elliptic operators [Tay13]. The principal symbols of the Laplacian are simply another means of expressing the Riemannian metric on the manifolds in question.

This completes the proof of the Theorem. □

3.1 Discussion of Corollaries

With $\{M_0^{i,j,k}\}$ and $\{M_1^{i,j,k}\}$ representing the two triple-product sets for the bases $\{e_0^i\}$ and $\{e_1^i\}$, let $z_i \in U_1$ be the U_1^∞ action on such an orthonormal basis $\{e_1^i\}$. Thus, we will choose z_i so that $\{z_i e_1^i\}$ yields $\{M_1^{i,j,k}\} = \{z_i z_j \bar{z}_k M_0^{i,j,k}\}$.

Why is this the case? In general, the symmetry group acting on the space of possible orthonormal bases of eigenfunctions is the space of Unitary Operators $U : \mathcal{H} \rightarrow \mathcal{H}$ that commute with projections $P_{\mathcal{V}_\lambda}$ onto the finite-dimensional eigenspaces $\mathcal{V}_\lambda$ associated with each individual eigenvalue λ of the Laplacian. Therefore

$$\begin{aligned} P_{\mathcal{V}_\lambda} U(e^i) &= U P_{\mathcal{V}_\lambda}(e^i), \therefore U(e^i) = \sum_{\lambda_i = \lambda_j} u_{ij} e^j \implies \\ M_U^{i,j,k} &:= \int_M U(e^i) U(e^j) \bar{U}(\bar{e}^k) \sqrt{g} dx = \sum_{\lambda_r = \lambda_i, \lambda_s = \lambda_j, \lambda_t = \lambda_k} u_{ir} u_{js} \bar{u}_{tk} M^{r,s,t} \end{aligned} \quad (12)$$

is the image of $M^{i,j,k}$ under U 's basis action $e^i \mapsto U(e^i)$.

Now under the conditions of Corollary 3, each of the $\mathcal{V}_\lambda$ are one dimensional vector spaces over $\mathbb{C}$, but that also means they are one dimensional vector spaces over $\mathbb{R}$, and so the full multiplicative symmetry group is $O(1, \mathbb{R})^\infty = \mathbb{Z}_2^\infty$.

Without the multiplicity constraint, the Corollary 3's associated prerequisite "regarding agreement in absolute values" would hence become "preservation of the ordered set of singular values of $\{M^{i,j,k}\}$, counted with multiplicity, when viewed as a collection of maps from $\mathcal{V}_{\lambda_i} \rightarrow \text{Hom}(\mathcal{V}_{\lambda_j}, \mathcal{V}_{\lambda_k}^*)$ ". It turns out this is a reasonably robust set of Unitary Invariants. We are significantly less confident that this generalized conjecture holds true, since it may be possible to produce a counterexample via explicit Sunada construction.

Getting back to the Corollary 2, we observe that the proof involves establishing this implication:

$$z_k = M_0^{i,\bar{i},k} / M_1^{i,\bar{i},k} \quad \forall i, k \in \mathbb{N}, \ni M_0^{i,\bar{i},k} \neq 0 \implies \exists r, s, t \in \mathbb{N} \ni \frac{M_0^{i,j,k}}{M_1^{i,j,k}} = \frac{M_0^{r,\bar{r},i} M_0^{s,\bar{s},j} M_0^{t,\bar{t},k}}{M_1^{r,\bar{r},i} M_1^{s,\bar{s},j} M_1^{t,\bar{t},k}}. \quad (13)$$

We may hope that for any given $k > 0$, $M_0^{i,i,k}$ cannot be identically 0 for all i , since it is a generically true condition, but false for specific cases like the flat tori case covered in the Example below. Furthermore, the Formula 13 for z_k requires both i -independence, and sufficiency, to establish the basis map $e_0^i \mapsto z_i e_1^i$ preserves $\{M_0^{i,j,k}\}$.

We sketch a proof of Corollary 2 (sufficiency) below the next set of formulae.

Nevertheless, let us compute some relevant identities so some intrepid future researcher can dig into this generalized conjecture:

$$\Delta f g = f \Delta g + g \Delta f - 2df \cdot dg \implies$$

$$M^{i,j,k} = 2 \frac{\langle de^i \cdot de^j | e^k \rangle}{\lambda_i + \lambda_j - \lambda_k}$$

Now by polarization

$$M^{i,j,k} = \frac{\langle (e^i + e^j)^2 - (e^i - e^j)^2 | e^k \rangle}{4} = \frac{M^{i,i,k} + M^{j,j,k} - \langle (e^i - e^j)^2 | e^k \rangle}{2},$$

and so the quadratic form

$$Q_k(f, g) := \langle df \cdot dg | e^k \rangle = \sum_{i,j} \hat{f}(i) \hat{g}(j) \langle de^i \cdot de^j | e^k \rangle$$

$$= \frac{1}{2} \sum_{i,j} \hat{f}(i) \hat{g}(j) (\lambda_i + \lambda_j - \lambda_k) M^{i,j,k}.$$

Now with J real-analytic

$$Q_k^J(f, g) := -\frac{1}{2} \langle (J(\sqrt{\Delta})fg - fJ(\sqrt{\Delta})g - gJ(\sqrt{\Delta})f | e^k \rangle$$

$$= -\frac{1}{2} (\langle fg | J(\sqrt{\Delta})e^k \rangle - \langle fJ(\sqrt{\Delta})g + gJ(\sqrt{\Delta})f | e^k \rangle)$$

$$= \frac{1}{2} \sum_{i,j} \hat{f}(i) \hat{g}(j) (J(\sqrt{\lambda_i}) + J(\sqrt{\lambda_j}) - J(\sqrt{\lambda_k})) M^{i,j,k} \quad (14)$$

$$\tilde{Q}_k(f, g) := -\frac{1}{2} \langle \sqrt{\Delta}fg - f\sqrt{\Delta}g - g\sqrt{\Delta}f | e^k \rangle$$

$$= \frac{1}{2} \sum_{i,j} \hat{f}(i) \hat{g}(j) (\sqrt{\lambda_i} + \sqrt{\lambda_j} - \sqrt{\lambda_k}) M^{i,j,k}$$

$$df \cdot dg = \sum_k Q_k(f, g) e^k = -\frac{\Delta fg - f\Delta g - g\Delta f}{2}$$

$$Q_0(f, f) = \frac{1}{\sqrt{\text{vol}(M)}} \sum_i \hat{f}(i)^2 \lambda_i$$

$$df \cdot df = \sum_k Q_k(f, f) e^k = \frac{1}{2} \sum_{i,j,k} \hat{f}(i) \hat{f}(j) (\lambda_i + \lambda_j - \lambda_k) M^{i,j,k} e^k$$

$$= \frac{1}{4} \sum_{i,j,k} \hat{f}(i) \hat{f}(j) (\lambda_i + \lambda_j - \lambda_k) (M^{i,i,k} + M^{j,j,k} - \langle (e^i - e^j)^2 | e^k \rangle) e^k$$

$$= g^2 = \sum_{i,j,k} \hat{g}(i) \hat{g}(j) M^{i,j,k} e^k \implies$$

$$\frac{1}{2} \sum_{i,j} \hat{f}(i) \hat{f}(j) (\lambda_i + \lambda_j - \lambda_k) M^{i,j,k} = \sum_{i,j} \hat{g}(i) \hat{g}(j) M^{i,j,k}$$

$$= \widehat{g^2}(k).$$

Note: for the one-dimensional flat-tori case below, $\tilde{Q}_k(e^i, e^j) = 0$ since $\sqrt{\Delta} = \sqrt{-1} \frac{d}{dx}$ is a true derivation.

4 Proof of Corollaries

Proof. Now consider the famous associativity relations from Conformal Field Theory:

$$\begin{aligned}
e^i e^j e^k &= \sum_{\ell} \langle e^i e^j | e^{\bar{k}} e^{\ell} \rangle e^{\ell} = \sum_{\ell, r} M^{i, j, r} \bar{M}^{\bar{k}, \ell, r} e^{\ell} \\
&= \sum_{\ell, r} M^{i, k, r} \bar{M}^{\bar{j}, \ell, r} e^{\ell} \quad .
\end{aligned} \tag{15}$$

$i = \bar{j}, \ell = k$ and relabeling $\implies$

$$\sum_r M^{i, \bar{i}, r} \bar{M}^{j, \bar{j}, r} = \sum_r |M^{i, j, r}|^2$$

Corollary 2 follows from the fact that $z_k \in U_1$ is well-defined under its presumed hypotheses, and the prior observation that the algebraic operators defined by $z_i z_j \bar{z}_k M_0^{i, j, k}$ and $M_1^{i, j, k}$ are both associative, and agree with pointwise function multiplication by squares of absolute values of eigenfunctions the generated algebra of which is also dense in $L^2(M, g)$ (again by Stone-Weierstrass, since it is presumed to separate points in the chosen target basis). Establishing $\det M^{i, \bar{i}, k} \neq 0$ infinitely often on finite-dimensional submatrices would suffice for this purpose.

Why? Let $\mathcal{V}$ be the closure of the vector space generated by $\{|e_i|^2\}$, and $\mathcal{W}$ for $\{e^k\}$. Then for any given $x \in M$, the commutative diagram

$$\begin{array}{ccc}
M & \xrightarrow{1} & M \\
\{ |e^i|^2(x) \} \downarrow & & \downarrow \{ e^k(x) \} \\
\mathbb{C}^{\mathbb{N}} & \xrightarrow{[M^{i, \bar{i}, k}]} & \mathbb{C}^{\mathbb{N}}
\end{array} \tag{16}$$

and the infinitely-often, finite-dimensional bijectivity of the infinite-dimensional unitary operator $[M^{i, \bar{i}, k}] : \mathcal{V} \rightarrow \mathcal{W}$ implies the evaluation map $\{|e^i|^2(x)\}$ is bijective (since bijectivity known for the other evaluation map $\{e^k(x)\}$), so it separates points by definition.

This completes the proof of Corollary 2.

$$\forall k > 0 \exists i > 0 \ni M^{i, \bar{i}, k} \neq 0. \tag{17}$$

Corollary 3 sufficiency follows by noting that the Equation 17 is generically true, as is the nonvanishing determinant condition in the prior paragraph. And if for some choice of i, j, k , the product $M^{i, \bar{i}, k} \bar{M}^{j, \bar{j}, k}$ disagreed between bases, it would disagree in every pair of bases.

Why? Since generic manifolds can be presumed to also have multiplicity-1 spectra, this reduces the full symmetry group to U_1^{∞} where these products are invariants, contradicting Theorem 1. This establishes the necessity of the hypothesis, and completes the proof. $\square$

5 Example

Let $\{\lambda_i\} \subset \mathbb{R}^n$ be an indexed, rank n lattice of Lie Algebra weights for the quotient space representation of $\mathfrak{g} = \mathbb{R}^n$ as translation invariant (i.e., constant) vector fields on itself, when $\mathbb{R}^n$ is also viewed as $\mathfrak{g}$'s associated Lie Group over a torus defined by $\mathbb{R}^n / AZ^n$, $A \in GL(n, \mathbb{R})$. These weights define integrable lifts of 1-forms over the torus that integrate to linear functionals $\langle x | \lambda_i \rangle$, $x \in \mathbb{R}^n$ as its Lie Group (covering the torus). These linear functionals can then be uniformly rescaled (by $2\pi\sqrt{-1}$) and exponentiated to form multiplicative characters that descend to form an orthonormal basis of $L^2(\mathbb{R}^n / AZ^n, dx)$, with Lebesgue (Haar) measure dx .

Moreover, this basis simultaneously diagonalizes the flat torus's Laplacian, because the Laplacian is the image of a symmetric, negative-definite quadratic Casimir element under this (constant coefficient linear differential operator) quotient space representation of the universal enveloping algebra. Hence, its eigenvalues are in constant proportion (of $4\pi^2$) to the Casimir-element-determined-length-squared of each character's weight in the lattice.

We presently view the above basis

$$\{e^{2\pi\sqrt{-1}\langle x|\lambda_i\rangle}/\sqrt{|\det A|}\}_{i=0}^\infty \quad (18)$$

to be our Theorem-applicable Fourier basis of orthonormal (multiplicative character) eigenfunctions (of this quotient representation of the (negative) Euclidean Casimir element) directly corresponding to $\{\lambda_i\}$. By our Theorem's hypotheses, we must have $i < j \implies \|\lambda_i\| \leq \|\lambda_j\|$ (with the Euclidean norm on the weights).

Now we can compute

$$M^{i,j,k} = \begin{cases} 1/\sqrt{|\det A|} & \lambda_i + \lambda_j - \lambda_k = 0 \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

As this Equation 19 is only invariant under linear transformations on the weight lattice $(A^{-1})^t\mathbb{Z}^n = \{\lambda_i\}$, only an L^2 orthonormal eigenfunction basis map which is induced from a volume-preserving invertible linear map between two such indexed, rank n weight lattices will keep the "algebraic/topological" indexed data set $\{M^{i,j,k}\}$ invariant.

However, in order to apply Theorem 1, it is essential that such a linear map B be $B \in SO(n, \mathbb{R})$ on the weight lattice, because the induced L^2 eigenfunction basis map

$$\{e^{2\pi\sqrt{-1}\langle x|B\lambda_i\rangle}/\sqrt{|\det A|}\}_{i=0}^\infty \quad (20)$$

must also preserve the "analytic" invariants – the Casimir-element induced figure $4\pi^2\|\lambda_i\|^2$ for each indexed weight, i.e. the individual eigenvalues of the flat-tori's Laplacian.

This representation-theoretical account [Kna01] is exactly equivalent to the prior development of *lattice congruence* [NRR23] traditionally used to delineate isometry classes of flat tori. In fact, the matrix transpose of such a linear map $B \in SO(n, \mathbb{R})$, as described in the prior paragraph, is the contravariant Riemannian isometry between the tori, as provided by application of the *Gelfand-Naimark Representation Theorem* during the proof of Theorem 1.

Acknowledgements The original research was funded in part by a gracious James Simons Research Award in 1995-1996, and the generous support of an Alfred P. Sloan Dissertation Fellowship in 1996-1997 at the University at Stony Brook.

The author would also like to thank Tanya Christiansen, Carolyn Gordon, Hamid Hezari, Harish Seshadri, and especially Leon Takhtajan for their technical assistance and review in the preparation of this manuscript for publication.

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